

The exponential function, its properties and graph

The variable moves into the index — and everything about the shape follows from whether the base is bigger or smaller than 1.

LESSON 49–50

2 × 40 MIN

ALGEBRA 10, §3.1

P2 · 2.1

LESSON CLOCK

10'

22'

20'

22'

16'

10'

The definition	10 min
Two shapes	22 min
The properties	20 min
Growth and decay	22 min
Practice	16 min
Homework	10 min

LEARNING OBJECTIVES

- State the definition and conditions on the base of an exponential function.
- Sketch $y = a^x$ for $a > 1$ and for $0 < a < 1$.
- State the domain, range, monotonicity and asymptote.
- Model exponential growth and decay.

TEXTBOOK

National	pp. 185–196
Cambridge	Pure Mathematics 2 & 3, pp. 26–32
Workbook	P2 Exercise 2A

01

Explanation

The definition

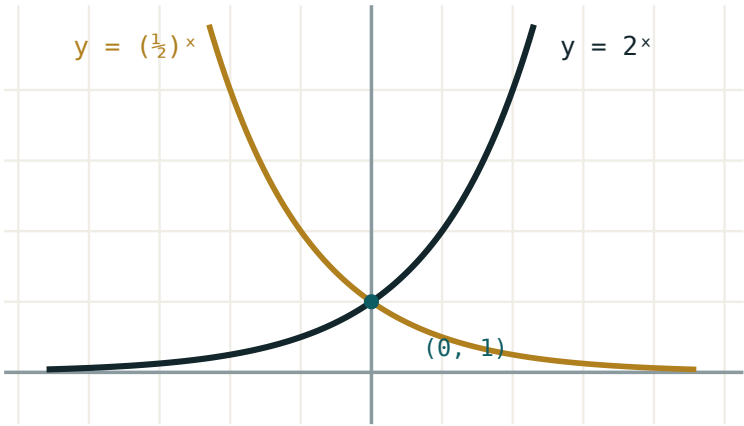
$$y = a^x, a > 0, a \neq 1$$

WHY THE BASE IS RESTRICTED

$a > 0$: otherwise $(-4)^{1/2}$ has no real value, and the function has holes. $a \neq 1$: $1^x = 1$ for every x — a constant, not an exponential.

The variable is in the **index**. That is the whole difference from a power function $y = x^n$, and it changes everything: 2^x eventually overtakes x^{100} .

Two shapes



both pass through $(0, 1)$; the x -axis is an asymptote

Growth when the base exceeds 1, decay when it is between 0 and 1. Both pass through $(0, 1)$.

PROPERTY	$A > 1$	$0 < A < 1$
domain	$\mathbb{R}$	$\mathbb{R}$
range	$y > 0$	$y > 0$
behaviour	increasing	decreasing
passes through	$(0, 1)$	$(0, 1)$
asymptote	$y = 0$	$y = 0$
as $x \rightarrow +\infty$	$y \rightarrow \infty$	$y \rightarrow 0$

THE GRAPH NEVER TOUCHES THE x -AXIS

a^x is positive for every real x . The range is $y > 0$, not $y \geq 0$ — and that is why every exponential equation can be divided by a^x safely.

The properties, and why they matter

Every one of these is used in the next four lessons:

- 1. **Monotonic**, so **one-to-one** — hence $a^m = a^n \Rightarrow m = n$, the whole method of solving exponential equations.
- 2. **Positive**, so $a^x = -3$ has no solution and $a^x > 0$ is always true.
- 3. **Increasing for $a > 1$** , so $a^m > a^n \Leftrightarrow m > n$; but for $0 < a < 1$ the inequality **reverses**.
- 4. **Invertible**, and its inverse is the logarithm of Lesson 57–58.

$$a^{(m+n)} = a^m a^n \cdot a^{(m-n)} = \frac{a^m}{a^n} \cdot (a^m)^n = a^{(mn)} \cdot a^0 = 1$$

Growth and decay

$$N(t) = N_0 \cdot a^t \quad a > 1 \text{ growth, } 0 < a < 1 \text{ decay}$$

SITUATION	MODEL	MEANING OF THE BASE
bacteria doubling hourly	$N = N_0 \cdot 2^t$	doubling time 1 hour
a 12% annual rise	$N = N_0 \cdot 1.12^t$	$1 + \frac{12}{100}$
a 7% annual fall	$N = N_0 \cdot 0.93^t$	$1 - \frac{7}{100}$
half-life h	$N = N_0 \cdot (\frac{1}{2})^{(t/h)}$	halving every h

Example. Uzbekistan's population grows about 1.7% a year. From 36 million, after 10 years: $36 \times 1.017^{10} \approx 42.6$ million.

02

Worked examples

EXAMPLE 1 Sketch $y = 3^x$ and state its domain, range and asymptote.

1

$a = 3 > 1$, so increasing.

2

Through $(0, 1)$ and $(1, 3)$.

3

Domain $\mathbb{R}$, range $y > 0$.

4

Asymptote $y = 0$.

ANSWER

Increasing; domain $\mathbb{R}$, range $y > 0$, asymptote $y = 0$

EXAMPLE 2 A colony of 500 bacteria doubles every 3 hours. Find the number after 12 hours.

$$① \quad N = 500 \times 2^{(t/3)}$$

$$② \quad t = 12 \Rightarrow 2^4 = 16$$

$$③ \quad 500 \times 16 = 8000$$

ANSWER

8000

EXAMPLE 3 A radioactive sample has half-life 5 days. What fraction remains after 20 days?

$$① \quad 20 \div 5 = 4 \text{ half-lives.}$$

$$② \quad \left(\frac{1}{2}\right)^4$$

$$③ \quad = \frac{1}{16}$$

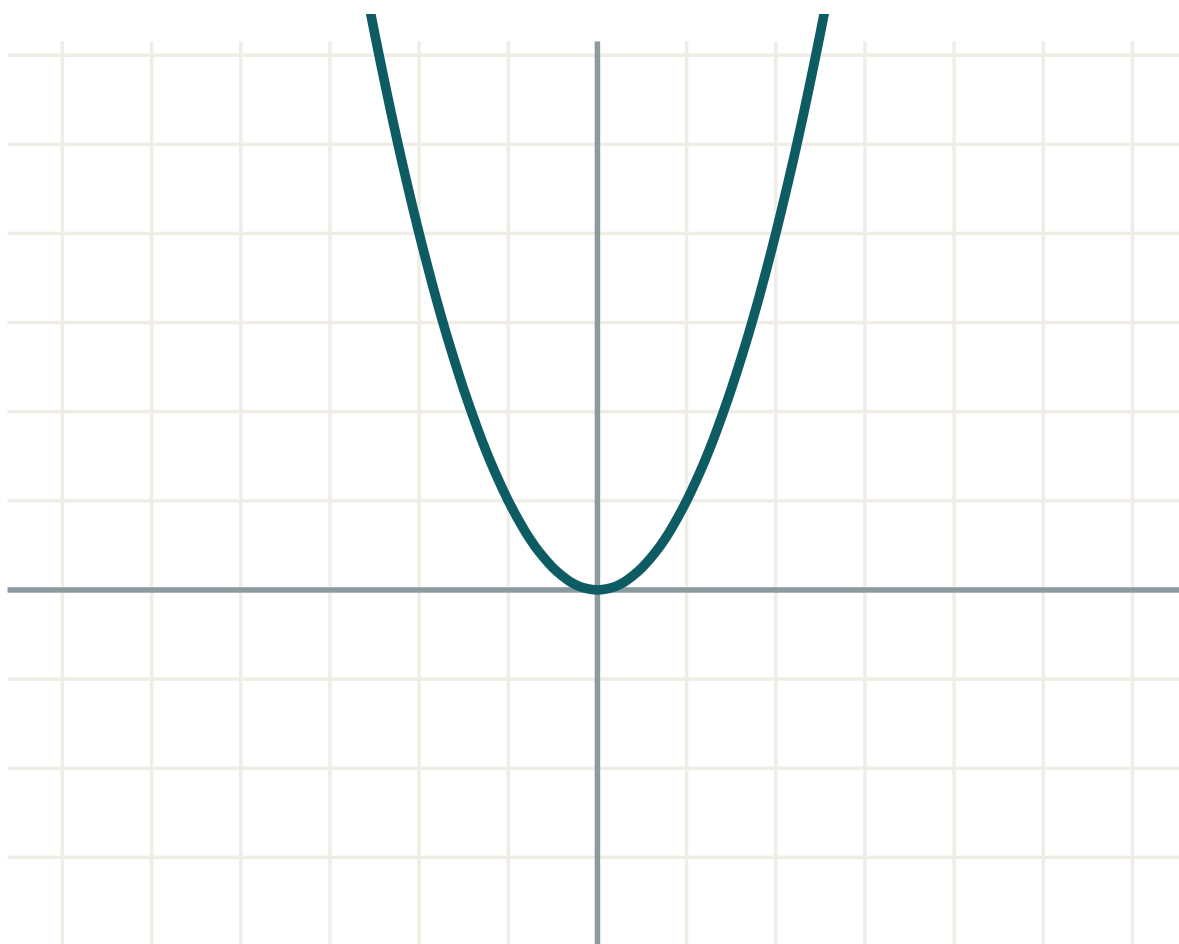
ANSWER

$\frac{1}{16}$ — about 6.25%

03 Interactive model

Change the base through 1 and watch the graph flip from growth to decay.

The exponential curve



a 1

b 0

c 0

equation $y = f(x)$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Exponential function	Ko'rsatkichli funksiya	Показательная функция
Base	Asos	Основание
Index (exponent)	Ko'rsatkich	Показатель
Growth	O'sish	Рост
Decay	Kamayish	Убывание
Horizontal asymptote	Gorizontal asimptota	Горизонтальная асимптота
Monotonic	Monoton	Монотонная
Doubling time	Ikkilanish vaqti	Время удвоения
Half-life	Yarim yemirilish davri	Период полураспада

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The range of $y = a^x$ is:

$y = a^x$ always passes through:

$y = 0.4^x$ is:

The base of an exponential must satisfy:

$$a > 0$$

$$a \neq 1$$

both

neither

A 5% annual fall gives base:

$$0.5$$

$$0.95$$

$$1.05$$

$$5$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Domain of $y = 2^x$

ANSWER $\mathbb{R}$

2. Range of $y = 2^x$

ANSWER $y > 0$

3. Is $y = 5^x$ increasing?

ANSWER yes

4. Is $y = 0.3^x$ increasing?

ANSWER no

5. 3^0

ANSWER 1

6. 2^5

ANSWER 32

7. $(\frac{1}{2})^3$

ANSWER $\frac{1}{8}$

Medium · the standard question

7 PROBLEMS

1. Base for a 12% annual rise

ANSWER 1.12

2. Base for a 7% annual fall

ANSWER 0.93

3. 500 bacteria doubling every 3 h: number after 12 h

ANSWER 8000

4. Half-life 5 days: fraction left after 20 days

ANSWER $\frac{1}{16}$

5. 36 million growing 1.7% a year: after 10 years

ANSWER ≈ 42.6 million

6. Asymptote of $y = 2^x + 3$

ANSWER $y = 3$

7. Range of $y = 2^x + 3$

ANSWER $y > 3$

Hard · stretch and reason

7 PROBLEMS

1. Sketch $y = 2^{x-1} - 4$: asymptote and intercepts

ANSWER asymptote $y = -4$, x -intercept 3

2. A car worth 200 million so'm falls 15% a year. Value after 4 years

ANSWER ≈ 104.4 million

3. When does $N = 1000 \times 1.05^t$ first exceed 2000?

ANSWER $t \approx 14.2$ years

4. A half-life of 8 days: what percentage remains after 12 days?

ANSWER $\approx 35.4\%$

5. Show 2^x eventually exceeds x^{10}

ANSWER Exponential growth beats any power

6. Solve $2^x = 8$ by inspection

ANSWER $x = 3$

7. Explain why $a^x = -2$ has no solution

ANSWER The range is $y > 0$

07 Homework

Homework — 5 tasks

Every sketch must show the asymptote as a dashed line.

- Sketch $y = 4^x$ and $y = (\frac{1}{4})^x$ on one set of axes, with domain, range and asymptote.
- Sketch $y = 3^x - 9$ and give its asymptote and both intercepts.
- A town of 40 000 grows 2.5% a year. Find its population after 8 years.
- A sample has half-life 6 hours. What fraction remains after 24 hours?
- Explain in three sentences why the base of an exponential function may not be 1.